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# APPENDICES

## Appendix A Calibration Curves for Gas Chromatography (Temperature of 45 °C, Retention Time of 60 Minute)

**Table A1** Calibration curve for hydrogen ($H_2$)

| Volume of Hydrogen (ml) | Peak Area |
|---|---|
| 0.02 | 16,313 |
| 0.04 | 58,770 |
| 0.08 | 180,674 |
| 0.1 | 226,743 |
| 0.2 | 427,198 |
| 0.4 | 778,509 |

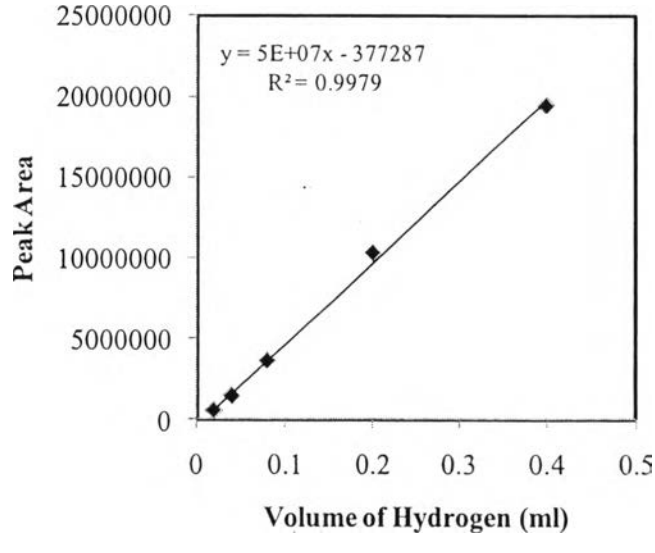



**Figure A1** The relationship between volume of hydrogen ($H_2$) and peak area.

**Equation**

$$\text{Amount of hydrogen} = \frac{\text{Peak area} + 377287}{5 \times 10^7} \qquad \text{(A1)}$$

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**Table A2** Calibration curve for nitrogen ($N_2$)

| Volume of Nitrogen (ml) | Peak Area |
|---|---|
| 0.02 | 34,210 |
| 0.04 | 128,767 |
| 0.08 | 305,287 |
| 0.1 | 393,916 |
| 0.2 | 809,433 |
| 0.4 | 1,602,475 |

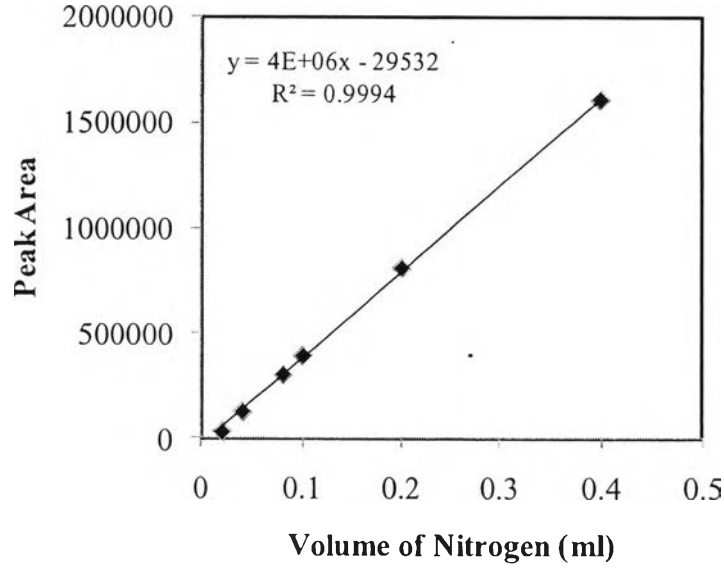



**Figure A2** The relationship between volume of nitrogen ($N_2$) and peak area.

**Equation**

$$\text{Amount of nitrogen} = \frac{\text{Peak area} + 29532}{4 \times 10^6} \quad \text{(A2)}$$

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**Table A3** Calibration curve for oxygen ($O_2$)

| Volume of Oxygen (ml) | Peak Area |
|---|---|
| 0.02 | 12,286 |
| 0.04 | 43,995 |
| 0.08 | 104,342 |
| 0.1 | 135,546 |
| 0.2 | 280,220 |
| 0.4 | 562,001 |

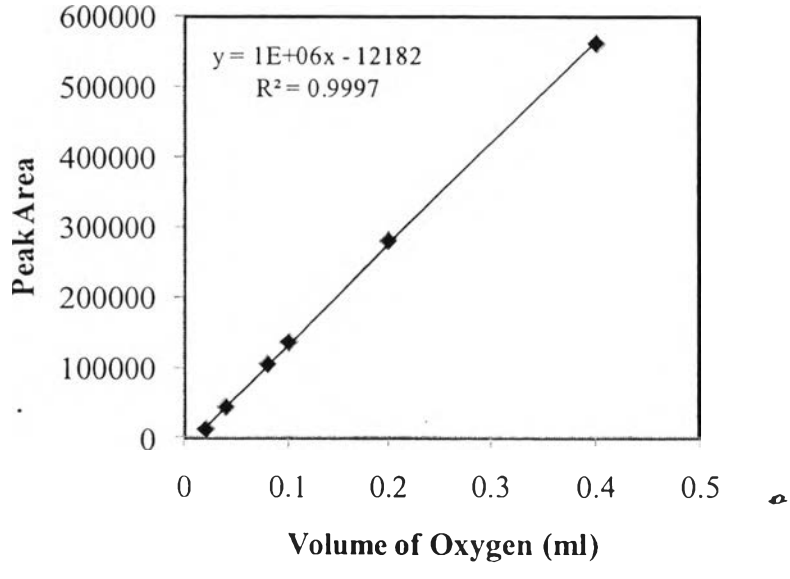



**Figure A3** The relationship between volume of oxygen ($O_2$) and peak area.

**Equation**

$$\text{Amount of oxygen} = \frac{\text{Peak area} + 12182}{1 \times 10^6} \qquad \text{(A3)}$$

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**Table A4** Calibration curve for methane ($CH_4$)

| Volume of Methane (ml) | Peak Area |
|---|---|
| 0.02 | 92,517 |
| 0.04 | 381,106 |
| 0.1 | 1,293,552 |
| 0.2 | 2,674,654 |

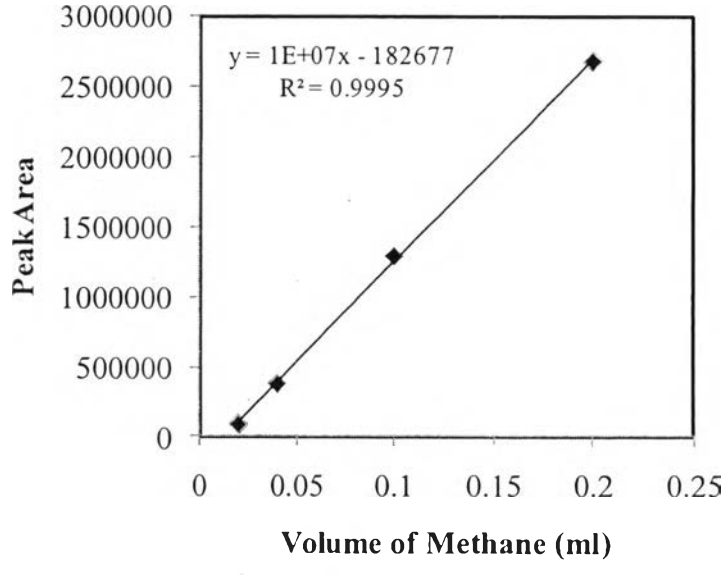



**Figure A4** The relationship between volume of methane ($CH_4$) and peak area.

**Equation**

$$\text{Amount of methane} = \frac{\text{Peak area} + 182677}{1 \times 10^7} \quad \text{(A4)}$$

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**Table A5** Calibration curve for carbon dioxide ($CO_2$)

| Volume of Carbon Dioxide (ml) | Peak Area |
|---|---|
| 0.02 | 26,118 |
| 0.04 | 97,539 |
| 0.08 | 220,122 |
| 0.2 | 596,414 |
| 0.4 | 1,315,885 |

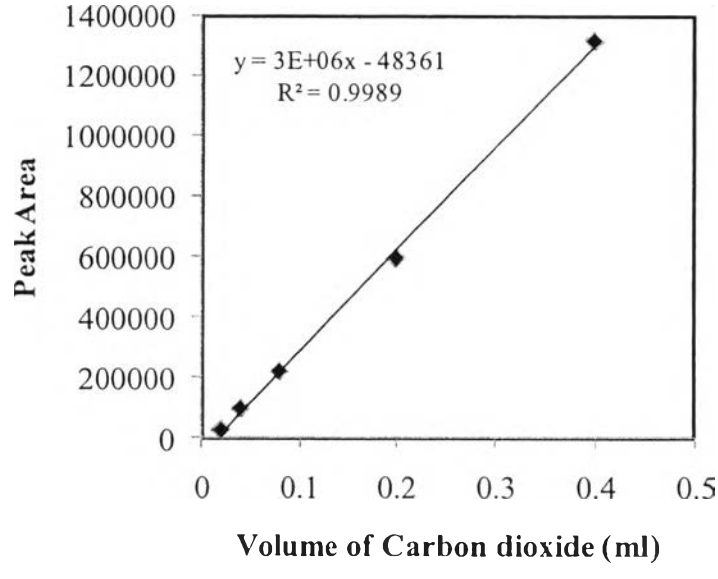



**Figure A5** The relationship between volume of carbon dioxide ($CO_2$) and peak area.

**Equation**

$$\text{Amount of carbon dioxide} = \frac{\text{Peak area} + 48361}{3 \times 10^6} \quad \text{(A5)}$$

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**Appendix B Calibration Curves for High Performance Liquid Chromatography (Temperatureof 45 °C, RetentionTimeof 60 Minute, Mobile Phase of 4 mM $H_2SO_4$)**

**Table B1** Calibration curve for ethanol ($C_2H_5OH$)

| Concentration of Ethanol (ppm) | Peak Area |
|---|---|
| 1000 | 189,866 |
| 2000 | 377,275 |
| 3000 | 569,223 |
| 4000 | 765,786 |
| 5000 | 958,108 |

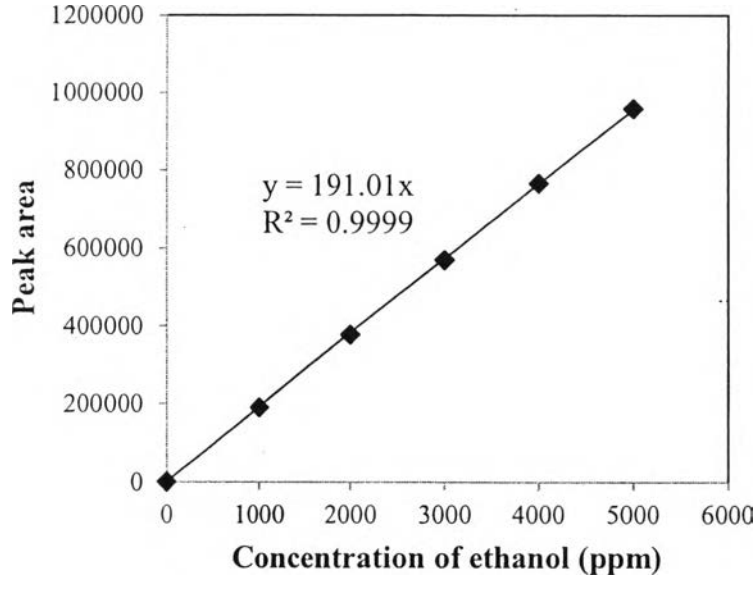



**Figure B1** The relationship between concentration of ethanol ($C_2H_5OH$) and peak area.

**Equation**

$$\text{Concentration of ethanol} = \frac{\text{Peak area}}{191.01} \qquad \text{(B1)}$$

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**Table B2** Calibration curve for lactic acid ($C_3H_6O_3$)

| Concentration of Lactic Acid (ppm) | Peak Area |
|---|---|
| 1000 | 241,057 |
| 2000 | 513,754 |
| 3000 | 735,330 |
| 4000 | 1,047,749 |
| 5000 | 1,350,707 |

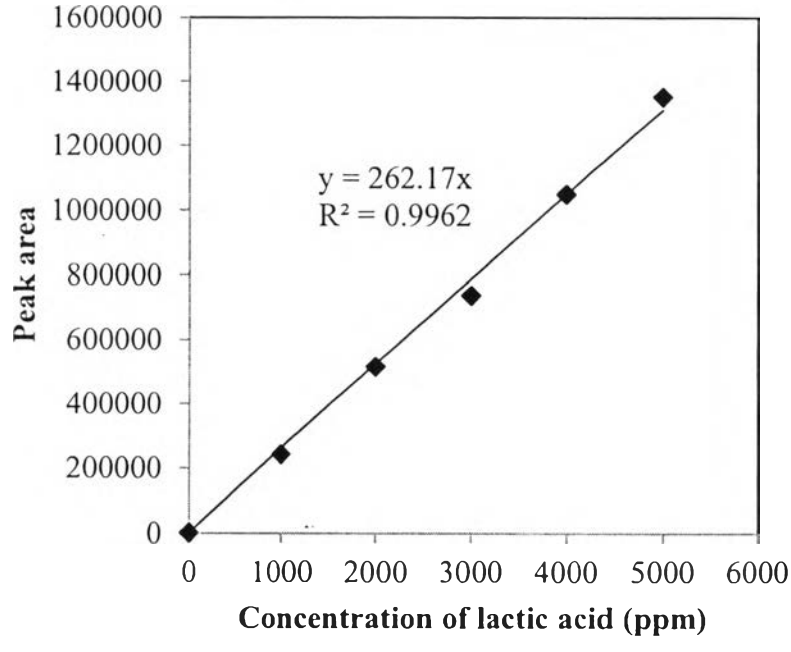



**Figure B2** The relationship between concentrations of lactic acid ($C_3H_6O_3$) and peak area.

**Equation**

$$\text{Concentration of lactic acid} = \frac{\text{Peak area}}{262.17} \qquad \text{(B2)}$$

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**Table B3** Calibration curve for valeric acid ($C_5H_{10}O_2$)

| Concentration of Valeric Acid (ppm) | Peak Area |
|---|---|
| 1000 | 346,808 |
| 2000 | 707,645 |
| 3000 | 108,2011 |
| 4000 | 1,470,955 |
| 5000 | 1,844,040 |

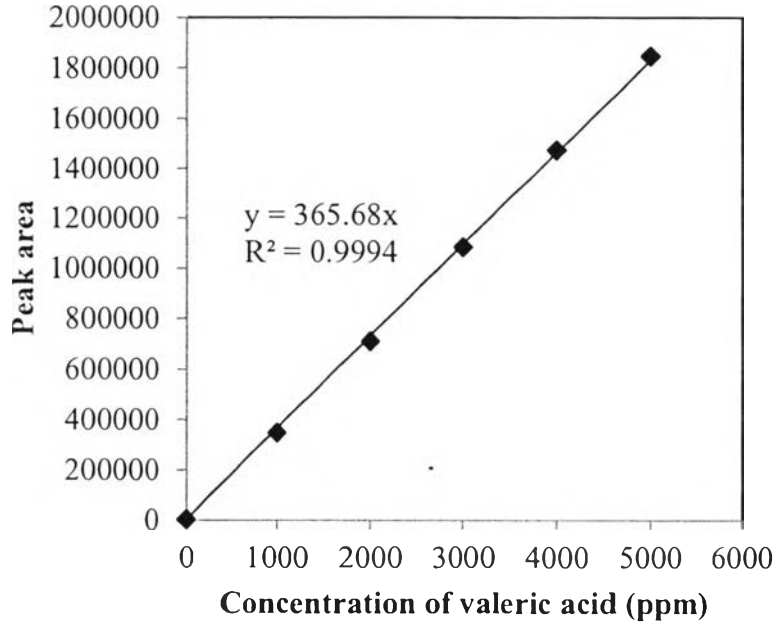



**Figure B3** The relationship between concentrations of valeric acid ($C_5H_{10}O_2$) and peak area.

**Equation**

$$\text{Concentration of valeric acid} = \frac{\text{Peak area}}{365.68} \qquad \text{(B3)}$$

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**Table B4** Calibration curve for propionic acid ($C_3H_6O_2$)

| Concentration of Propionic Acid (ppm) | Peak Area |
|---|---|
| 1000 | 274,670 |
| 2000 | 553,990 |
| 3000 | 836,683 |
| 4000 | 1,091,859 |
| 5000 | 1,435,669 |

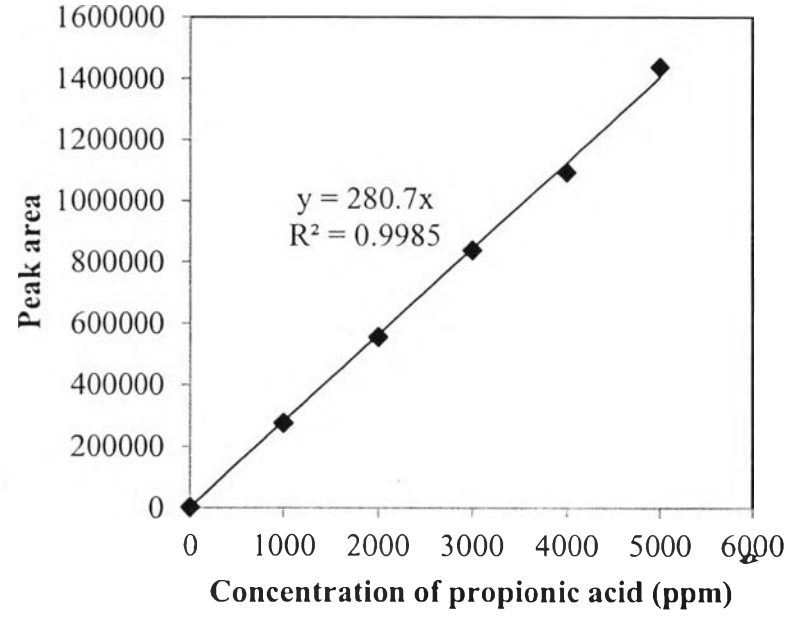



**Figure B4** The relationship between concentrations of propionic acid ($C_3H_6O_2$) and peak area.

**Equation**

$$\text{Concentration of propionic acid} = \frac{\text{Peak area}}{280.7} \qquad \text{(B4)}$$

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**Table B5** Calibration curve for acetic acid ($CH_3COOH$)

| Concentration of Acetic Acid (ppm) | Peak Area |
|---|---|
| 1000 | 226,593 |
| 2000 | 458,639 |
| 3000 | 693,445 |
| 4000 | 951,778 |
| 5000 | 1,179,161 |

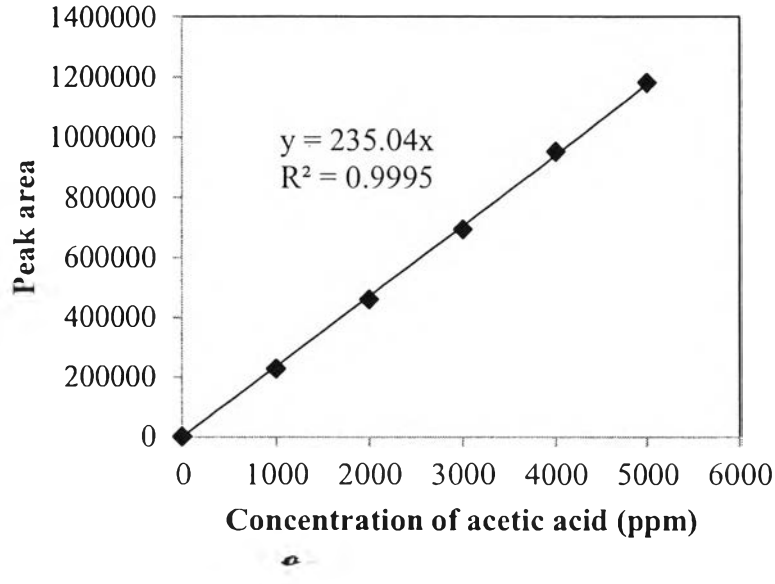



**Figure B5** The relationship between concentration of acetic acid ($CH_3COOH$) and peak area.

**Equation**

$$\text{Concentration of acetic acid} = \frac{\text{Peak area}}{235.04} \qquad \text{(B5)}$$

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**Table B6** Calibration curve for butyric acid ($C_3H_6O_3$)

| Concentration of Butyric Acid (ppm) | Peak Area |
|---|---|
| 1000 | 310,185 |
| 2000 | 636,623 |
| 3000 | 974,830 |
| 4000 | 1,315,752 |
| 5000 | 1,672,791 |

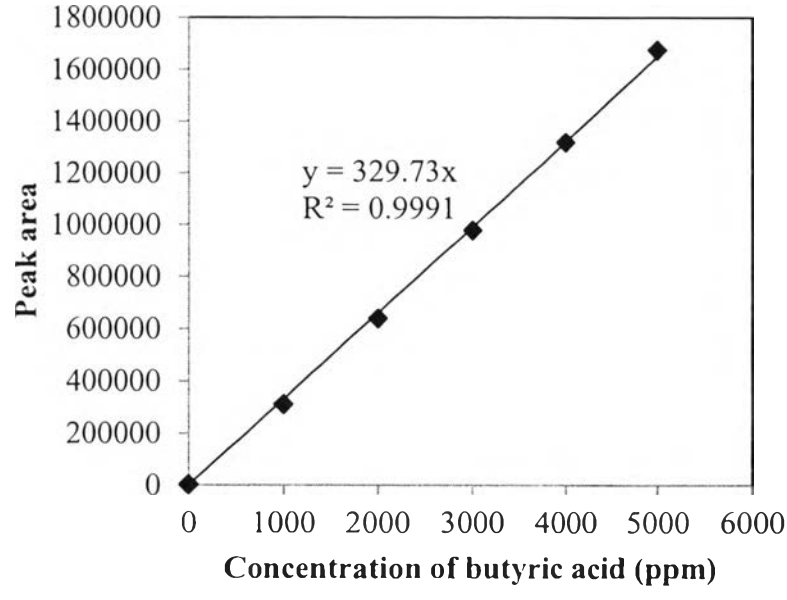



**Figure B6** The relationship between concentrations of butyric acid ($C_3H_6O_3$) and peak area.

**Equation**

$$\text{Concentration of butyric acid} = \frac{\text{Peak area}}{329.73} \qquad \text{(B6)}$$

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## Appendix C Preparation of 1 M NaOH Solution for pH-Controlled System

Preparation of NaOH at concentration of 1 M NaOH

$$= \frac{40 \text{ g NaOH}}{1 \text{ litter } H_2O} \quad \text{(C1)}$$

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**Proceedings**

1. Sri-ubon, N.; Chavadej, S.; and Intanoo, P. (2015, April 21) Enhancement of separate hydrogen and methane production from ethanol wastewater in a two-stage UASB unit by microaeration. Proceeding of The 6th Research Symposium on Petrochemical and Materials Technology and the 21st PPC Symposium on Petroleum, Petrochemicals and Polymers, Bangkok, Thailand.
2. Sri-ubon, N.; Chavadej, S.; and Intanoo, P. (2015, July 20-22) Optimization of separate hydrogen and methane production from alcohol wastewater in a two-stage UASB system. Proceeding of The International Conference on Chemical and Biochemical Engineering 2015, Paris, France.

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